AoMA-Optimized Scheduling for Energy-Constrained UAV Communication in Remote Surveillance

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Remote Surveillance in Poorly Connected Areas

Context

- Remote and rural sites often have weak or no communication infrastructure.
- Safety-critical alarms (fire, gas leak, intrusions) still need to reach operators.

Problem

- Outages and long delays can leave critical alarms unnoticed or unresolved.
- We focus on remote surveillance where alarms from distant sites require timely handling.

UAVs: Flexible but Energy-Constrained

Opportunity: UAV-Assisted Monitoring

- UAVs can act as flying relays or base stations above remote sites.
- They can be rapidly deployed and repositioned where alarms occur.

Challenge: Limited Onboard Energy

- Batteries tightly limit flight time, sensing, and communication.
- The UAV must balance clearing alarms and safely returning to charge.

Key question: How to schedule UAV actions under energy and safety constraints?

From AoI to AoMA (Semantic Freshness)

Classical Age of Information (AoI)

- AoI measures how fresh the received status information is.
- It treats all updates as equally important, regardless of their impact.

Semantic Focus: Age of Missed Alarm (AoMA)

- In alarm systems, missing a true alarm is far worse than slightly stale normal readings.
- AoMA measures how long true alarms remain unresolved and thus captures *risk*.

This thesis: optimize UAV scheduling to **minimize AoMA** in remote surveillance.

Research Questions

This thesis addresses the following key research questions:

- **RQ1 (Optimization)**: How can UAV scheduling be optimized to minimize the **Age of Missed Alarm (AoMA)** in remote monitoring systems?
- **RQ2 (Constraints)**: How can safety and energy constraints be incorporated into a **DRL framework** to prevent mission failure due to depletion or excessive delay?
- **RQ3 (Evaluation)**: How does the proposed DRL-based scheduling method perform compared to traditional **heuristic approaches** in terms of AoMA reduction and energy efficiency?

System Model: Environment

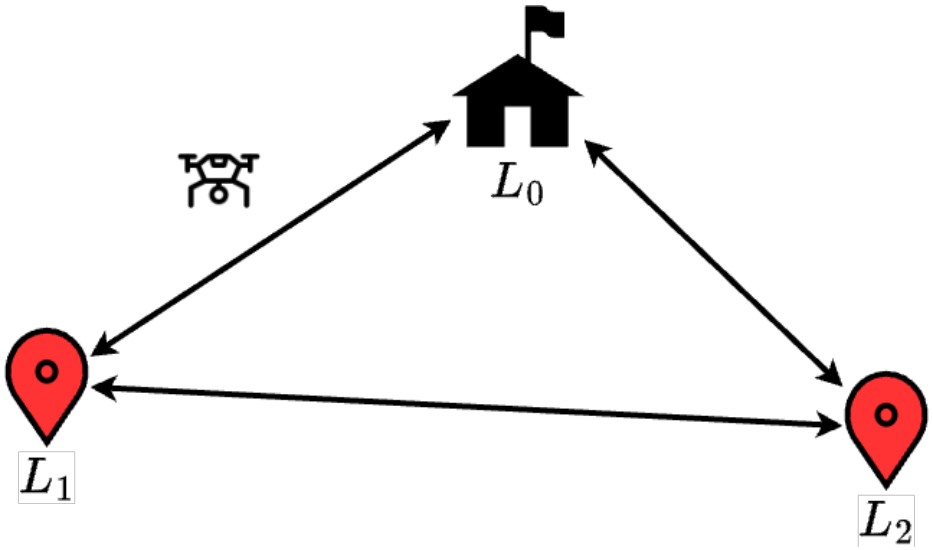


Figure: Syetem environment

We consider a single UAV with limited battery capacity operating over three locations: L_0, L_1, and L_2. L_0 is the home base where the UAV can recharge, while L_1 and L_2 are remote sites that generate alarms and need to be served.

System Model: Alarm status transition

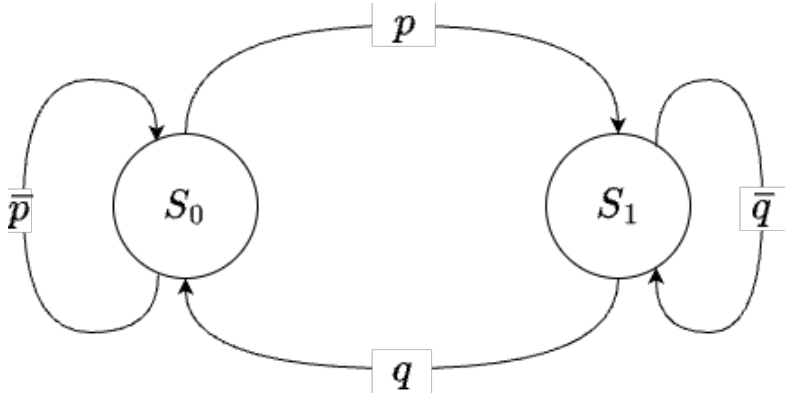


Figure: Alarm status transition model

Each remote site has an independent two-state Markov alarm process: S_0 is the safe state and S_1 is the alarm state. Transitions from S_0 to S_1 occur with probability p, while in this thesis we assume that alarms do not disappear automatically ($q = 1$) and are only cleared when the UAV visits the site. When a location is in S_1, its AoMA increases until the UAV clears the alarm.

System Model: AoMA in a Time-Slotted System

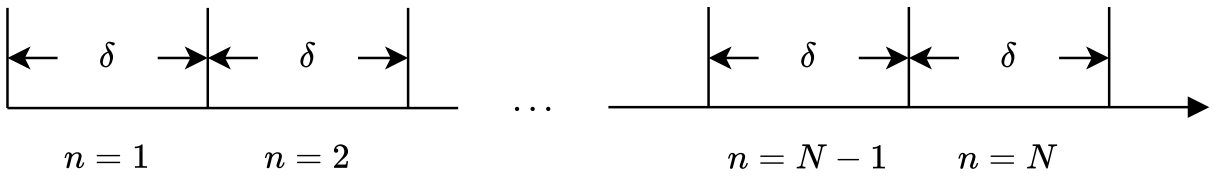


Figure: Time-slotted operation of the UAV with slot length δ.

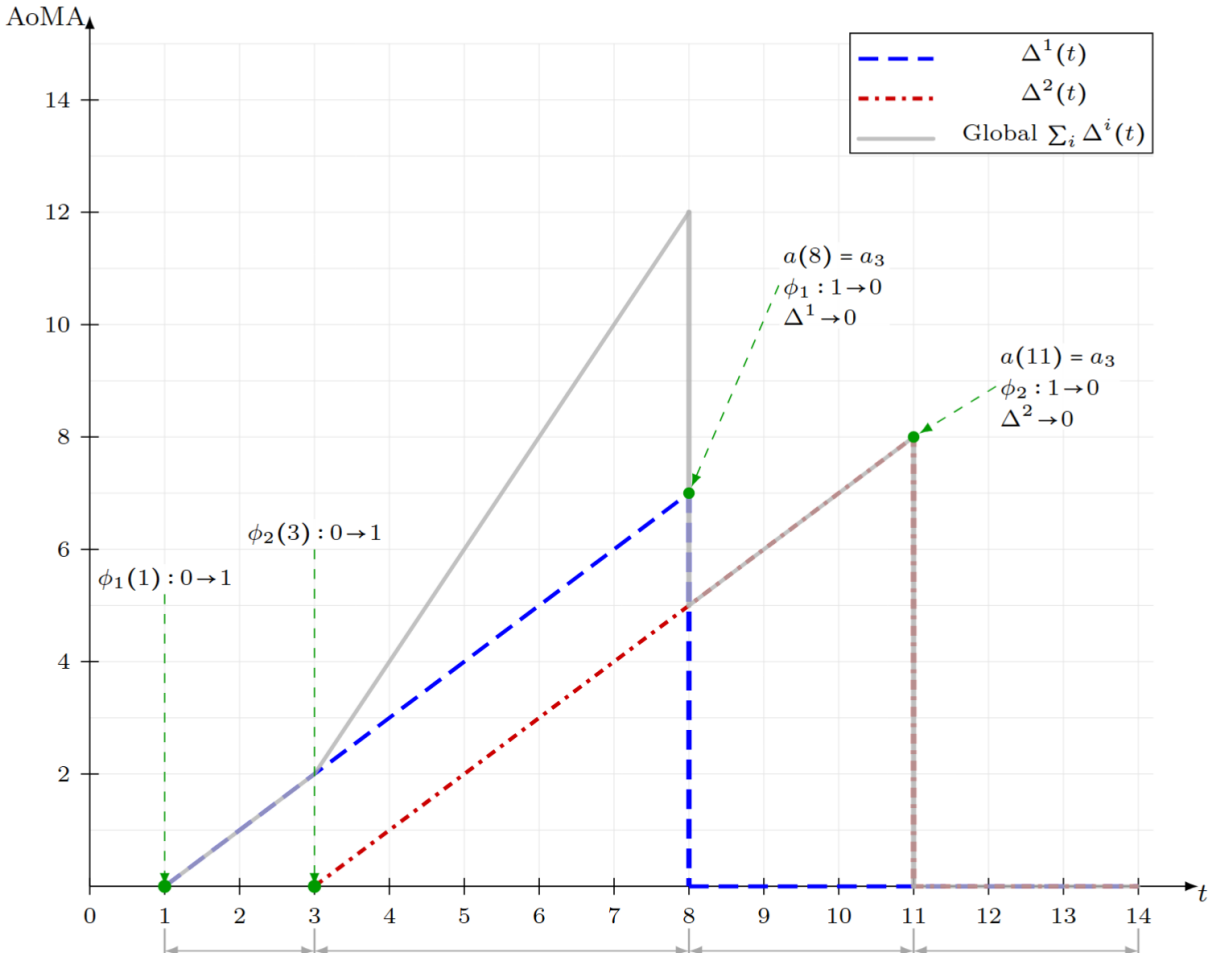
Per-location AoMA
For each location L_i and time slot n, the Age of Missed Alarm $\Delta^i(n)$ describes how long a true alarm at L_i has been active and not yet cleared. Given the alarm indicator $\phi_i(n) \in \{0, 1\}$,

$$\Delta^i(n+1) = \begin{cases} \Delta^i(n) + 1, & \text{if } \phi_i(n) = 1, \\ 0, & \text{if } \phi_i(n) = 0. \end{cases}$$

Global AoMA
The global AoMA at time slot n aggregates all locations:

$$\Delta_{\text{tot}}(n) = \sum_i \Delta^i(n).$$

System Model: AoMA Dynamics in a Two-Location Scenario



Problem Formulation: AoMA-Optimal Scheduling

Objective: minimize weighted average AoMA

$$J\big(\{a(n)\}\big) = \frac{1}{N}\sum_{n=0}^{N-1}\sum_{i=1}^{2} w_i\, f\big(\Delta_i(n)\big)$$

Subject to

$$0 \le b(n) \le E_{\max}, \quad n = 1, \ldots, N,$$
$$\Delta_i(n) \le \Delta_{\max}, \quad i \in \{1, 2\},\ n = 1, \ldots, N.$$

- $w_i > 0$: importance weight of alarms at location L_i.
- $f(\cdot)$: non-decreasing penalty function of AoMA.
- $\Delta_{\max}$: maximum permissible AoMA, enforcing timely alarm resolution.

MDP Formulation: State Space

The UAV alarm scheduling problem is modeled as an MDP with state s_n at time slot n defined by:

- **UAV position** $x(n) \in \{L_0, L_1, L_2\}$: current location of the UAV.
- **Battery level** $b(n) \in [0, E_{\max}]$: remaining battery energy.
- **Alarm status** $\Phi(n) = \big(\phi_1(n), \phi_2(n)\big)$: alarm indicators at L_1 and L_2.
- **AoMA vector** $\Delta(n) = \big(\Delta_1(n), \Delta_2(n)\big)$: per-location AoMA values.

Thus, the full state at time slot n is

$$s_n = \big(x(n),\, b(n),\, \Phi(n),\, \Delta(n)\big).$$

MDP Formulation: Action Space

At each time slot n, the UAV selects one action $a(n) \in \mathcal{A} = \{a_0, a_1, a_2, a_3, a_4\}$:

$$a(n) = \begin{cases} a_0 : \text{navigate to } L_0, \\ a_1 : \text{navigate to } L_1, \\ a_2 : \text{navigate to } L_2, \\ a_3 : \text{clear alarm at current location}, \\ a_4 : \text{wait at current location}. \end{cases}$$

Each action a_j incurs an energy cost c_j, $j \in \{0, 1, 2, 3, 4\}$, with navigation costs proportional to the distance between the current and target locations.

MDP Formulation: Reward Function

The overall reward r_n guides the agent toward minimizing AoMA while respecting energy and safety constraints.

Reward decomposition

$$r_n = \underbrace{r_n^{\text{alarm}}}_{\text{Latency penalty}} + \underbrace{r_n^{\text{clear}}}_{\text{Clearance reward}} + \underbrace{r_n^{\text{energy}}}_{\text{Energy shaping}} + \underbrace{r_n^{\text{penalty}}}_{\text{Safety penalty}} .$$

1. Alarm penalty (minimize AoMA):

$$r_n^{\text{alarm}} = -\sum_{i=1}^{2} \left(\alpha_1 + \alpha_2 \frac{f(\Delta_i(n))}{\Delta_{\max}} \right)$$

2. Clearance reward (incentive):

$$r_n^{\text{clear}} = \sum_{i=1}^{2} \mathbb{I}_{i,n}\, w_i \max\left\{ R_{\min},\, R_{\max} - \frac{f(\Delta_i(n))}{\beta} \right\}$$

where $\mathbb{I}_{i,n} = 1$ if an alarm at L_i is cleared in slot n, and 0 otherwise.

3. Energy shaping (soft constraint):

$$r_n^{\text{energy}} = \begin{cases} -(E_{\max} - b(n)) + \rho, & b(n) \le \rho, \\ 0, & \text{otherwise.} \end{cases}$$

4. Safety penalty (hard constraint):

$$r_n^{\text{penalty}} = \begin{cases} -p_{\text{anchor}}, & b(n) < 0, \\ -p_{\max}, & \max_i \Delta_i(n) > \Delta_{\max}. \end{cases}$$

DRL Solution: Deep Reinforcement Learning (DDQN)

Why Double DQN?

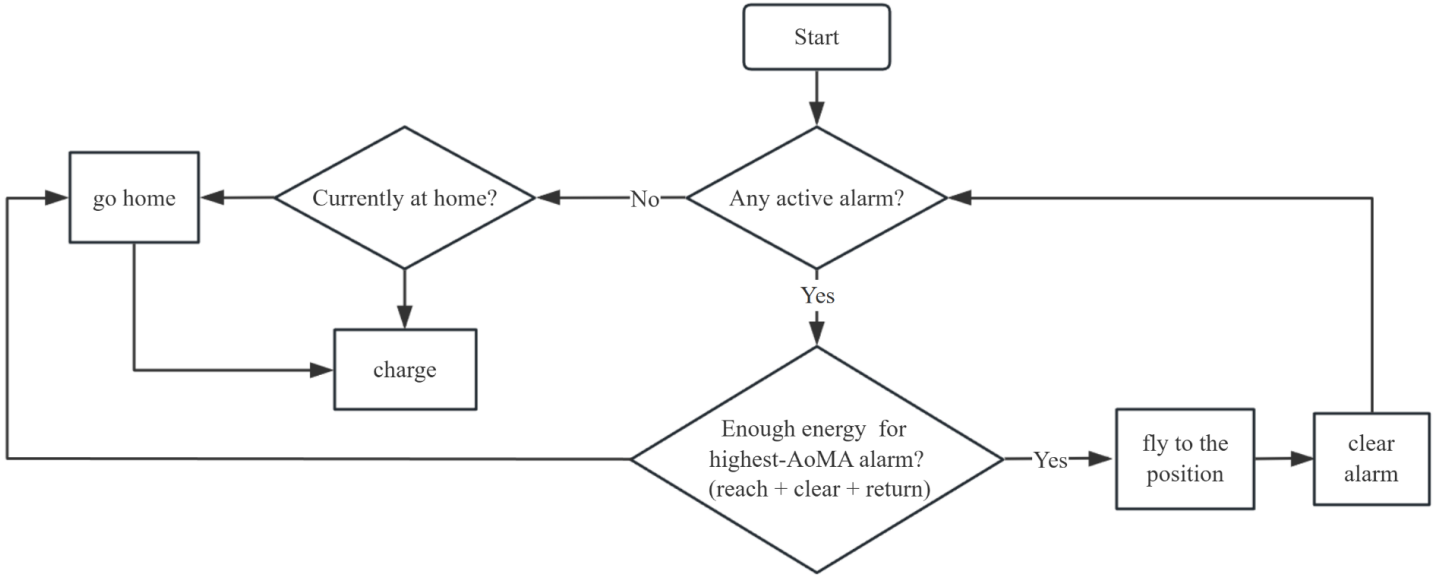
Standard DQN suffers from maximization bias (value overestimation). We adopt **DDQN** to decouple action selection from target evaluation, ensuring stable learning convergence.

Key Mechanisms for AoMA Optimization:

- **Long-term Planning**: Unlike myopic heuristics, the agent optimizes cumulative future rewards, learning to sacrifice immediate small gains for long-term AoMA minimization.
- **Experience Replay**: Stores transitions (s, a, r, s') in a buffer to break temporal correlations and improve sample efficiency.

Baseline Solution: Greedy Heuristic

Strategy: A deterministic policy that always selects the feasible alarm with the highest current AoMA.



Note: This heuristic serves as a **strong baseline** because it guarantees zero battery failures by design, although it lacks foresight.

Common Implementation Setup

1. System Environment

Based on System Model Parameters (Table 4.2):

Parameter	Value
Battery Capacity	$E_{\max} = 80$
Charging Rate	$r_c = 20$
Clearance Cost	$c_3 = 5$
Max AoMA Threshold	$\Delta_{\max} = \mathbf{120}$
Episode Length	$T_{sim} = 10000$

Topology Distances:

- $d(L_0, L_1) = 10, d(L_0, L_2) = 20$
- $d(L_1, L_2) = 15$

2. DDQN Agent Setup

Based on Training Parameters (Table 4.1):

- **Network Architecture**:
  - Input Dim: 6 (State features)
  - Hidden Layers: **[128, 64, 32]**
  - Output Dim: 4 (Actions)

Hyperparameter	Value
Learning Rate	$\alpha = 2 \times 10^{-4}$
Discount Factor	$\gamma = \mathbf{0.96}$
Replay Buffer	50000
Batch Size	64
Target Sync (C)	200 steps
Exploration Decay	0.997

Scenario 1: Alarm-Intensive Environment

Scenario Characteristics:

- High alarm frequency.
- Frequent conflicts between patrolling and charging.

Specific Parameters:

Parameter	Value
Alarm Prob. (L_1)	$p_1 = \mathbf{17}\%$
Alarm Prob. (L_2)	$p_2 = \mathbf{12}\%$
Total Load	*Intensive*

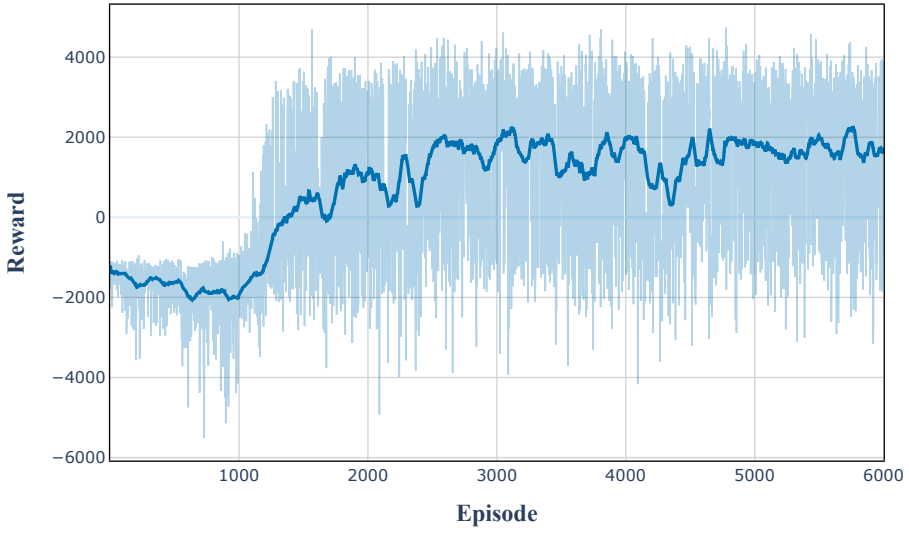


Figure: Training reward convergence of DDQN agent. The moving average (window=100) shows stable learning progress.

Scenario 1: Training Evolution Details

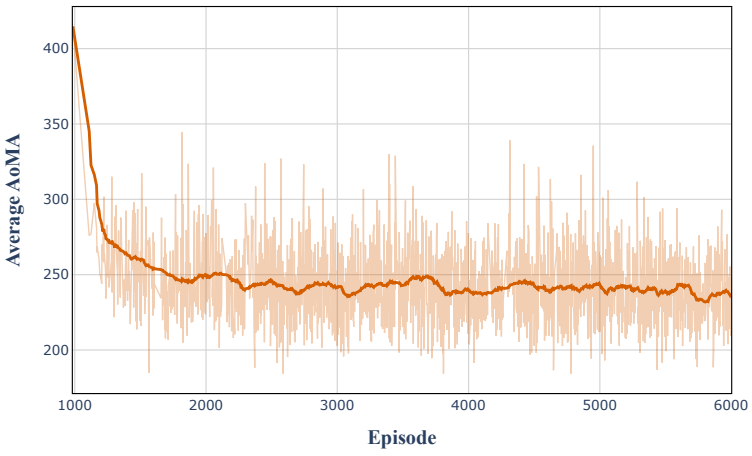


Figure: **AoMA Reduction**: The average AoMA decreases significantly as the agent learns to prioritize urgent alarms.

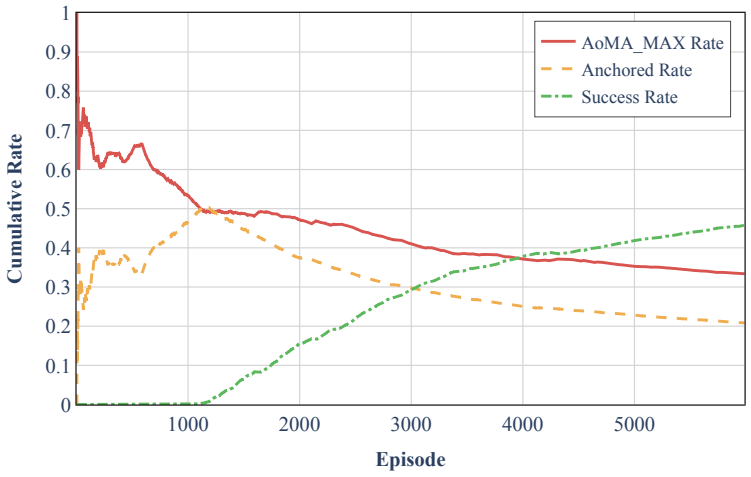


Figure: **Safety Analysis**: Success rate (green) increases while failure rates (red/orange) decline over episodes.

Scenario 1: Test Performance Comparison

Performance Statistics:

Metric	DDQN	Greedy
Average AoMA	**221.89**	236.96
Maximal AoMA	**265.05**	274.55
Minimal AoMA	**183.00**	191.00
Failures	0	0

Analysis:

- DDQN achieves **lower AoMA** across all metrics.
- Proactive scheduling reduces peak delays.

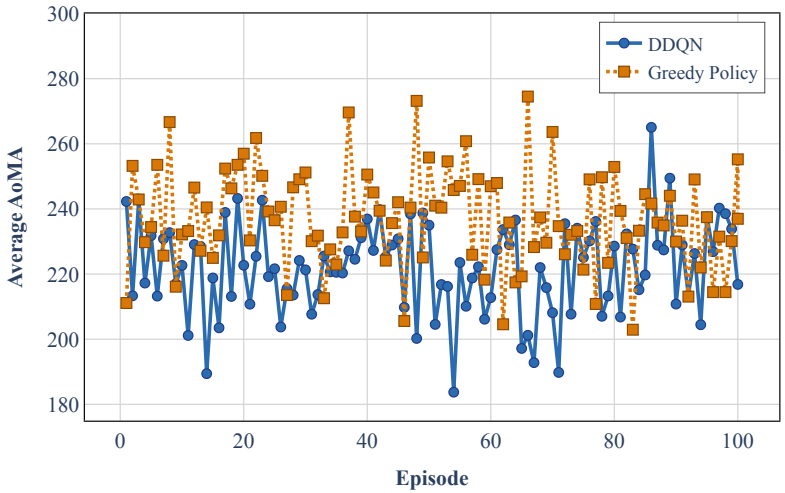


Figure: Average AoMA over 100 test episodes. DDQN consistently outperforms the Greedy baseline in this intensive scenario.

Scenario 2: Alarm-Sparse Environment

Scenario Characteristics:

- Low alarm frequency.
- Long idle periods requiring energy conservation.

Specific Parameters:

Parameter	Value
Alarm Prob. (L_1)	$p_1 = \mathbf{4}\%$
Alarm Prob. (L_2)	$p_2 = \mathbf{8}\%$
Total Load	*Sparse*

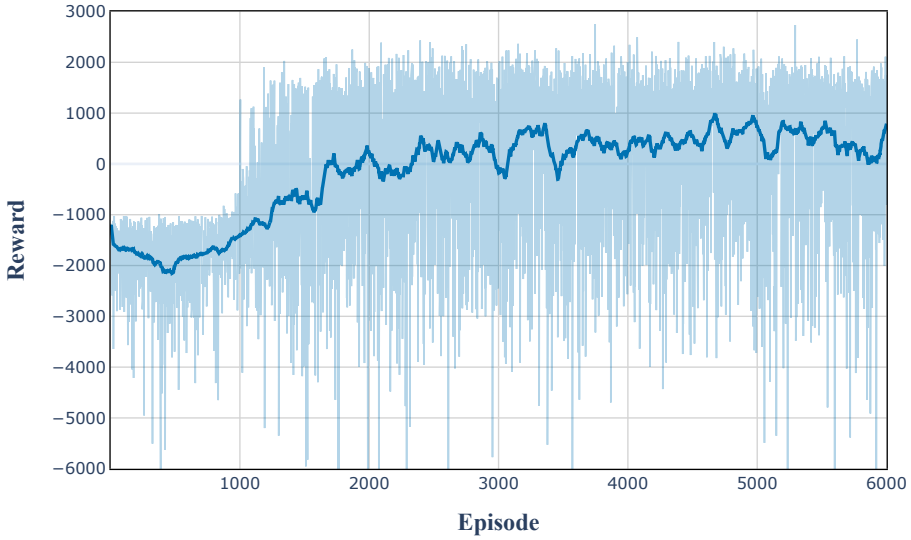


Figure: Training reward convergence. Note the faster convergence and smaller fluctuations compared to Scenario 1 due to lower task difficulty.

Scenario 2: Training Evolution Details

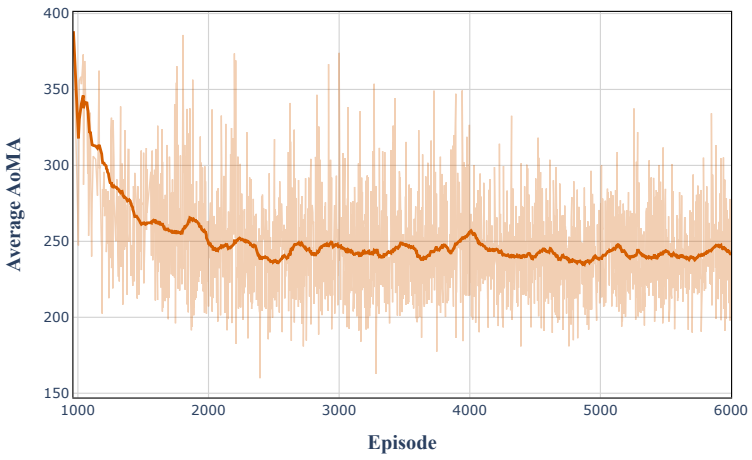


Figure: **AoMA Reduction**: A sustained downward trend is observed. Early data points are sparse as only successful episodes are plotted.

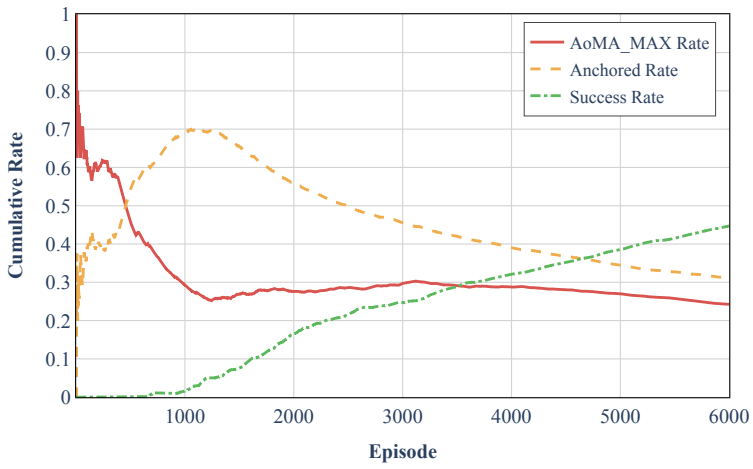


Figure: **Safety Analysis**: Unlike Scenario 1, *anchored failures* (orange) dominate early training, while AoMA violations (red) are rare.

Scenario 2: Test Performance Comparison

Performance Statistics:

Metric	DDQN	Greedy
Average AoMA	234.47	**229.83**
Maximal AoMA	279.50	**271.74**
Minimal AoMA	**185.63**	189.45
Failures	0	0

Analysis:

- Greedy policy is slightly more stable in sparse settings.
- DDQN shows higher variance (lower min, higher max) due to exploration.

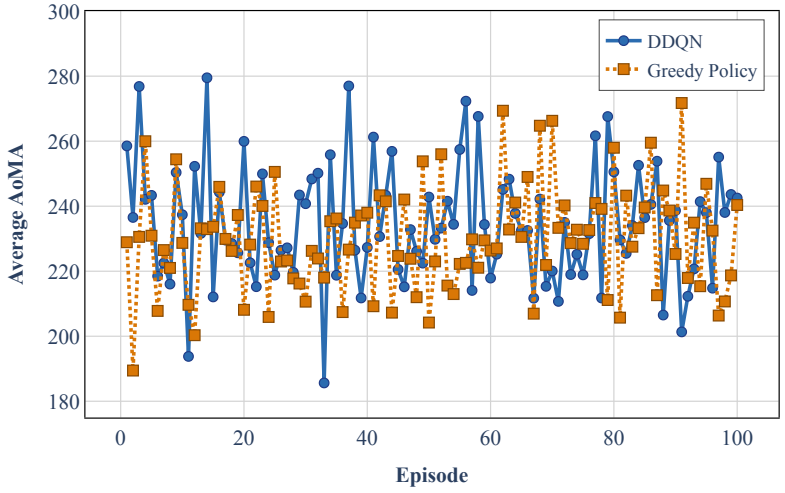


Figure: Average AoMA over 100 test episodes. The performance gap is negligible, indicating Greedy is sufficient for low-load tasks.

Answers to Research Questions

- **RQ1: AoMA Optimization via DRL**
  - While DRL cannot guarantee global optimality, it effectively learns **high-quality policies** that significantly reduce long-term AoMA compared to myopic heuristics, particularly in complex, alarm-dense environments.
- **RQ2: Incorporating Safety & Energy Constraints**
  - Achieved through **Reward Shaping**:
  - Positive rewards encourage timely clearance, while **strict penalties** discourage battery depletion and AoMA violations, successfully guiding the agent toward energy-aware and safe behaviors.
- **RQ3: DRL vs. Greedy Baseline**
  - **Scenario 1 (Intensive)**: DDQN **outperforms** the greedy baseline in both AoMA reduction and energy efficiency.
  - **Scenario 2 (Sparse)**: Both methods achieve **comparable performance**. Given its deterministic nature and zero training cost, the Greedy policy remains a competitive choice for low-load tasks.

Conclusion and Future Directions

Summary of Contributions

- **Systematic Investigation**: Proposed a unified framework optimizing **AoMA** for remote surveillance, prioritizing semantic urgency.
- **Realistic Modeling**: Developed a simulation capturing stochastic alarms, energy dynamics, and multi-site constraints.
- **Algorithmic Success**: The **DDQN policy** proved robust, significantly outperforming the Greedy baseline in high-stress scenarios.

Future Work

- **Scalability**: Extending to **Multi-UAV** cooperation for load balancing.
- **Complexity**: Handling **Partial Observability** (POMDP) where alarm states are not instantly known.
- **Safety**: Integrating explicit safety constraints (e.g., Constrained MDPs) to guarantee zero-failure operations.

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